

Using Generating functions to solve recurrence Relation

Problem: 1- Solve the recurrence relation

$$a_r - 2a_{r-1} + a_{r-2} = 2^r \quad \text{for } r = 2, 3, 4 \dots \quad \text{and}$$

boundary condition $a_0 = 2$, and $a_1 = 1$

$$\Rightarrow a^r x^r - 2a_{r-1}x^r + a_{r-2}x^r = 2^r x^r$$

$$\sum_{r=2}^{\infty} a^r x^r - \sum_{r=2}^{\infty} 2a_{r-1}x^r + \sum_{r=2}^{\infty} a_{r-2}x^r = \sum_{r=2}^{\infty} 2^r x^r$$

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Problem: 2- Solve the recurrence relation

$$a_r - 5a_{r-1} + 6a_{r-2} = 2 \quad \text{for } r = 2, 3, 4 \dots \quad \text{and} \\ \text{initial condition } a_0 = 1, \text{ and } a_1 = 1$$

Problem: 3- Solve the recurrence relation

$$a_r - 8a_{r-1} = 10^{r-1} \quad \text{for } a_1 = 9$$

solve recurrence Relation

Problem: 1- Solve the recurrence relation

$$a_n = 3a_{n-1} + 2n$$

Problem: 2- Solve the recurrence relation

$$a_n - a_{n-1} - 2a_{n-2} = n^2$$